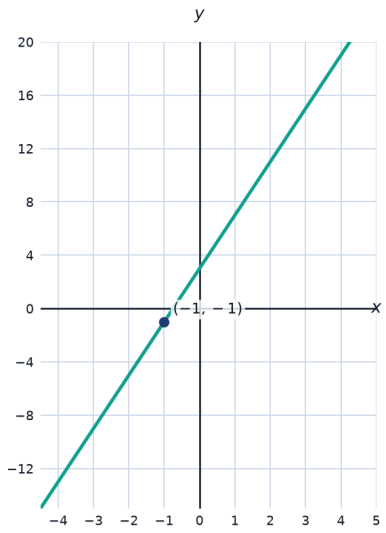
Exponential & Logarithmic Functions

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

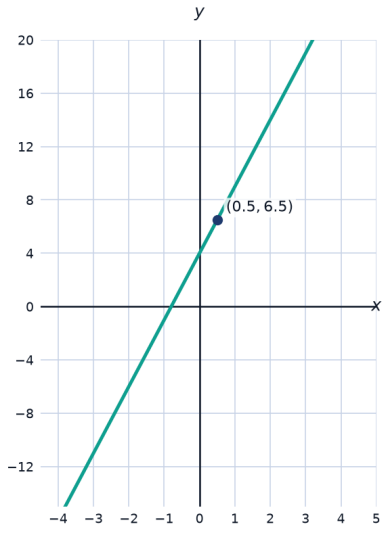
1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 4x + 3$ in negative exponents, evaluate $f(5)$.



2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 5x + 4$ in the laws of exponents and scientific notation, evaluate $f(2)$.



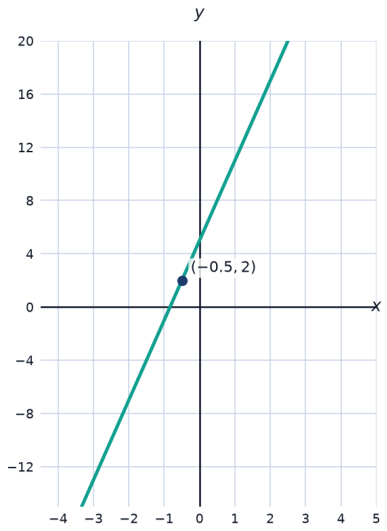
3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A data set recorded for the laws of exponents and scientific notation consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean? Show your complete algebraic work.

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4 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 6x + 5$ in patterns with rational and irrational numbers, evaluate $f(2)$.

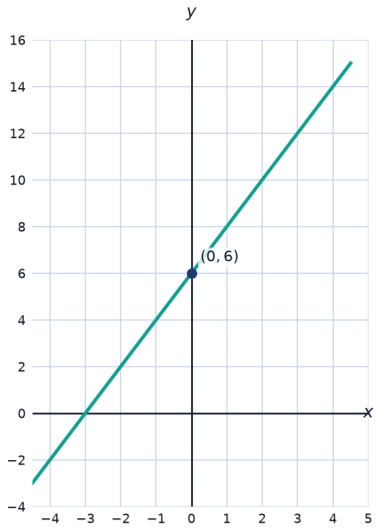


5 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A data set recorded for patterns with rational and irrational numbers consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean? Show your complete algebraic work.

6 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

For the model $f(x) = 2x + 6$ in simplifying radical expressions, evaluate $f(2)$.



7 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A data set recorded for simplifying radical expressions consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean? Show your complete algebraic work.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

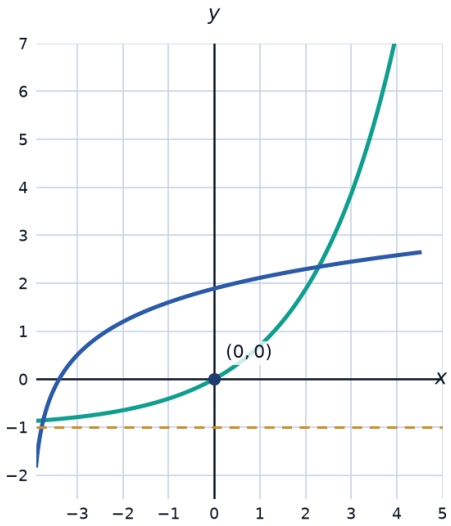
Solve $2^{x+1} = 5^{x-1}$, giving your answer to 3 significant figures.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Solve the logarithmic equation $\log_3(x^2 - 7) = 2$ for all real solutions, and verify that all roots lie within the valid domain.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Level 2 · MediumAn ecologist models the population P of trout in a lake as $P(t) = 1400 - 35t$, where t is the number of years after 2015. What is the best interpretation of the number 35 in this model?



11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Level 3 · AdvancedWhich ordered pair (x, y) is a solution to the system of inequalities $y > 2x - 3$ and $y \leq -x + 4$?

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Level 4 · StrategyA student finishes the first 18 questions of Module 1 with 9 minutes remaining. Questions 19 and 20 are lengthy geometry proofs, while Questions 21 and 22 are quick Desmos system intersection problems. In what order should the student attack the remaining 4 questions, and why?

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Level 2 · MediumA student solves a linear equation and obtains the fraction $x = -\frac{7}{4}$. What are the valid ways to enter this answer into the Digital SAT Student-Produced Response (SPR) grid?

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Level 1 · FoundationsFind the x-intercept and y-intercept of the linear equation $3x - 6y = 18$.

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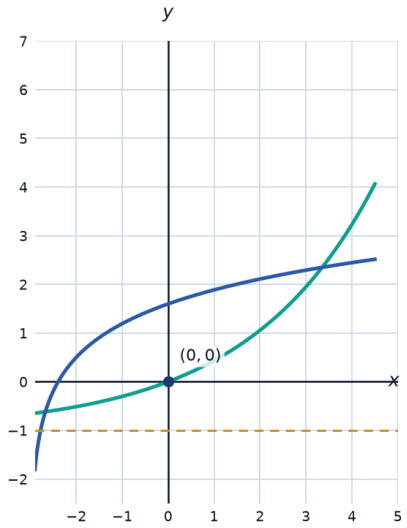
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Linear Equations & Lines

If $f(x) = 4x - 3$ and $g(x) = x^2 + 1$, find (a) $gf(2)$ and (b) an expression for $g^{-1}(x)$ for $x \geq 1$.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 10.01 Rational Expressions

Level 3 · AdvancedFor the piecewise function $g(x) = \{2x + 5, x \leq -1; x^2 - 3, x > -1$, what is the value of $g(-1) + g(3)$?



17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 11.01 Statistics & Data Analysis

Level 2 · MediumA study of 100 students recorded test results based on exam prep: • Studied: 54 passed, 6 failed (60 total) • Did not study: 18 passed, 22 failed (40 total) What is the probability that a randomly chosen student passed the exam, given that the student studied?

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Quadratic Functions

Level 4 · SynthesisIf $f(x - 3) = 4x + 5$ for all real numbers x, what is the value of $f(2)$?

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Exponents & Polynomials

Level 2 · MediumLet $f(x) = x - 4$ and $g(x) = 2x^2 + 1$. If $h(x) = g(f(x))$, what is the value of $h(1)$?

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.01 Functions & Transformations

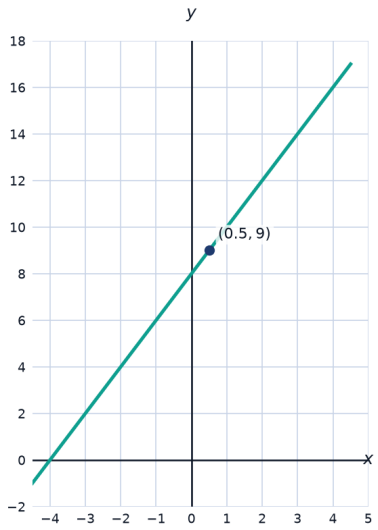
Level 1 · FoundationsFor the function $f(x) = 2x^2 - 3x + 5$, what is the value of $f(-3)$?

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 2x + 8$ in multiplication properties of exponents, evaluate $f(2)$.



a. 12

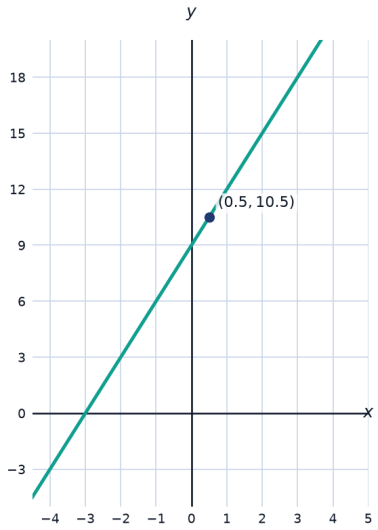
b. 14

c. 4

d. 16

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 9$ in division properties of exponents and the zero exponent, evaluate $f(2)$.



a. 15

b. 18

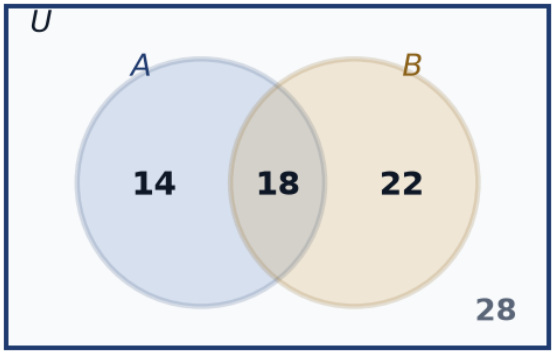
c. 6

d. 19

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for multiplication properties of exponents consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?



a. 18.5

b. 19.5

c. 20.5

d. 22.5

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for division properties of exponents and the zero exponent consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

a. 19.5

b. 20.5

c. 21.5

d. 23.5

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The function f is defined by $f(x) = a \cdot b^x + c$, where a, b, and c are constants with $a > 0$, $b > 0$, and $beq1$. In the xy-plane, the horizontal asymptote of the graph of $y = f(x)$ is the line $y = -4$. If the graph passes through the points $(0, -1)$ and $(2, 8)$, what is the value of $f(3)$?

a. 20

b. 24

c. 32

d. 44

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26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

The function $f(x) = a \cdot 2^x + k$ has a horizontal asymptote at $y = 3$ and passes through the point $(0, 7)$, where a and k are real constants. What is the value of $f(2)$?

a. 19

b. 31

c. 16

d. 11

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

What is the horizontal asymptote of the graph of the function $f(x) = 3 \cdot 2^{x+1} - 5$?

a. $y = -5$

b. $y = 0$

c. $y = 3$

d. $x = -1$

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The graph of the exponential function $f(x) = a \cdot b^x$ passes through the points $(1, 12)$ and $(3, 108)$. Determine the value of a and the value of b.

a. a = 4, b = 3

b. a = 3, b = 4

c. a = 6, b = 2

d. a = 2, b = 6

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

The graph of the exponential function $f(x) = a \cdot b^x$ passes through the points $(1, 12)$ and $(3, 108)$. Determine the value of a.

a. 2

b. 3

c. 4

d. 6

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the value of the exponential function $f(x) = 2^x$ when $x = 3$?

a. 6

b. 8

c. 9

d. 5

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the value of y when $x = 3$ for the function $f(x) = 2^{x+1}$?

a. 4

b. 8

c. 6

d. 16

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

What is the half-life formula for exponential decay?

a. $A(t) = A_0(2)^{t/h}$

b. $A(t) = A_0e^{-t}$

c. $A(t) = A_0/2^t$

d. $A(t) = A_0\left(\frac{1}{2}\right)^{t/h}$

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Solve: $e^{2x} = 5$. Express in terms of $\ln$.

a. $x = \ln k**2 + 1 - k$

b. $x = \ln \frac{k**2+1}{k}$

c. $x = \frac{\ln k**2+1}{k}$

d. $x = \frac{k}{\ln k**2+1}$

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

If compound interest is $A = P(1 + \frac{r}{n})^{nt}$, which formula gives continuous compounding?

a. $A = Pe^{rt}$

b. $A = P/e^{rt}$

c. $A = P(1 + r)^t$

d. $A = Prt$

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

The equation $\log_a x = b$ and $\log_b x = a$ simultaneously implies:

a. No simultaneous solution

b. $a = b$

c. $x = 0$

d. $x = a^b = b^a$

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36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Solve: $5^x = 12$. Give exact answer.

a. $x = 12/5$

b. $x = \frac{\ln 12}{\ln 5}$

c. $x = \ln 12 - \ln 5$

d. $x = \log_{12} 5$

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Write using natural log: $\log_b x =$

a. $\frac{\ln b}{\ln x}$

b. $\ln x - \ln b$

c. $\ln x \cdot \ln b$

d. $\frac{\ln x}{\ln b}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Quadratic Functions

The value of e to 4 decimal places is:

a. 2.1787

b. 3.1416

c. 2.6180

d. 2.7183

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Exponents & Polynomials

Convert to exponential form: $\log_5 25 = 2$.

a. $5^{25} = 2$

b. $2^5 = 25$

c. $5^2 = 25$

d. $25^2 = 5$

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.01 Functions & Transformations

Which is the y-intercept of $f(x) = 5 \cdot 3^x$?

a. $(1, 15)$

b. $(0, 15)$

c. $(0, 5)$

d. $(0, 3)$

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41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Systems & Inequalities

The doubling time model is: $A = A_0 \cdot 2^{t/d}$. If $d = 5$, find A/A_0 at $t = 10$.

a. 8

b. 2

c. 4

d. 10

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 7.01 Sequences & Series

Which of these is equivalent to $e^{2\ln 3}$?

a. $2e^3$

b. 9

c. 6

d. $\ln 9$

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 8.01 Factoring Polynomials

Find the derivative of $f(x) = x^x$.

a. $x^x \ln x$

b. x^{x+1}

c. $x \cdot x^{x-1}$

d. $x^x(1 + \ln x)$

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 9.01 Radicals & Exponents

Find the number of solutions to $e^x = x + 2$.

a. 1

b. 2

c. Infinitely many

d. 0

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.01 Linear Equations & Lines

Solve: $3^x = 2^{x+1}$. Use logarithms.

a. $x = \frac{\ln 2}{\ln 3}$

b. $x = 1$

c. $x = \frac{\ln 3}{\ln 2}$

d. $x = \frac{\ln 2}{\ln 3 - \ln 2}$

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